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Architecture Selection & Justification Document

Architecture Selection: "We chose **Layered Architecture** for the Self-Service Coffee Kiosk System."

Architectural Choice: We selected the **Layered Architecture** pattern (comprising Presentation, Business, and Data layers) because it provides a clear separation of concerns tailored to standalone hardware systems.

Two Scenario-Related Reasons:

1. **Separation of Hardware and Business Logic:** Isolating touchscreen UI hardware interactions from payment logic and backend databases simplifies maintenance. Updating receipt printer drivers or updating UI touchscreen layouts will not disrupt core order management logic.
2. **Low Latency for Standalone Hardware:** Because all operations occur on a single kiosk terminal, Layered Architecture eliminates unnecessary network overhead and deployment complexity associated with microservices, ensuring fast response times for customer interactions.

Security Advantage: The isolated Data Layer and Payment Service Component restrict payment processing data (credit card information) strictly within the Business Layer. The Presentation Layer cannot directly query menu databases or payment gateway credentials, preventing unauthorized database modifications or exposure of transaction logs via UI exploits.

Performance Benefit: Direct in-memory communication between layers minimizes invocation latency between the user interface, order processing module, and database, enabling instant menu rendering and immediate receipt output during peak hours.